

Unified Military Analytics and Comparison Dashboard

Complete Project Documentation (Milestone 1 – 4)

1. Project Overview

The DV Unified Military Analytics and Comparison Dashboard is an end-to-end data analytics system designed to collect, process, analyze, and visualize global military power data. The system integrates web scraping, data preprocessing, feature engineering, and interactive dashboards to allow users to explore defense capabilities of countries and simulate coalition strength.

The project follows a complete data pipeline lifecycle: Data Collection → Cleaning → Feature Engineering → Visualization → Validation → Deployment

The platform provides four analytical modules: 1. Quick Stats – Global ranking overview 2. Nation Overview – Detailed country analysis 3. Compare Powers – Country vs Country comparison 4. Coalition Builder – Multi-country alliance simulation

2. Milestone 1 — Data Collection and Preparation

2.1 Web Scraping

Military data was collected from publicly available defense statistics pages using Python.

Libraries Used: - requests - BeautifulSoup - pandas

Process: 1. Read country URLs from predefined list 2. Send HTTP request to each page 3. Parse HTML metric blocks 4. Extract numerical values 5. Store into structured dataset

Extracted Metrics Included: - Active personnel - Reserve personnel - Aircraft - Tanks - Naval fleet - Defense budget - Geographic indicators

Output: military_raw_data.csv

2.2 Data Cleaning

The raw scraped dataset contained textual numbers, symbols, and inconsistent formatting.

Cleaning Steps: - Removed commas, percentage symbols and plus signs - Converted strings to numeric types - Standardized column names - Handled missing values - Normalized inconsistent entries

Output: unified_military_data.csv

3. Milestone 2 — KPI Engineering and Dashboard Planning

3.1 Feature Engineering

New analytical indicators were derived to make comparisons meaningful.

KPIs Created: - Power Index Rank Gap - Assets Per Capita - Budget to GDP Ratio

The dataset was reshaped into formats usable for dashboard tools.

Output: unified_military_KPI.xlsx

3.2 Dashboard Design

Wireframes were designed before development to define user flow and interactions.

Planned Interactions: - Country filters - Region filters - Parameter selection - Cross-dashboard navigation

Prototype dashboard was created using sample data to validate layout logic.

4. Milestone 3 — Dashboard Development

The full analytics application was built using visualization software.

4.1 Quick Stats Dashboard

Purpose: Provide a global overview of military power distribution

Features: - Top countries by power index - Budget comparison charts - Global map visualization - Dynamic KPI cards

4.2 Nation Overview Dashboard

Purpose: Deep analysis of a selected country

Features: - Radar and bar charts - Personnel breakdown - Equipment counts - Defense spending indicators

4.3 Compare Powers Dashboard

Purpose: Side-by-side comparison between two countries

Features: - Parameter-based country selection - Metric comparison charts - Power index comparison - Geographic comparison maps

4.4 Coalition Builder Dashboard

Purpose: Evaluate combined strength of alliances

Features: - Multi-country selection - Aggregated totals calculation - Coalition vs country comparison - Strategic capability estimation

All dashboards were connected through navigation buttons and shared filters to ensure a unified experience.

5. Milestone 4 — Testing, Validation and Delivery

5.1 Functional Validation

All filters, parameters and navigation paths were tested to ensure correct data response.

5.2 Data Verification

Random country values were compared with processed dataset outputs to confirm accuracy of displayed metrics.

5.3 Performance Optimization

Dashboard responsiveness was tested with full dataset size and optimized to avoid lag or freezing.

5.4 Documentation and Packaging

The project repository was organized into scripts, data, dashboards and documentation sections for reproducibility and evaluation.

6. Technology Stack

Programming: Python Data Processing: pandas, numpy Web Scraping: requests, BeautifulSoup
Visualization: Power BI / Tableau Documentation: Markdown, GitHub

7. Final Outcome

The project successfully demonstrates a complete real-world data analytics workflow from data acquisition to decision-support visualization. The final dashboard enables users to analyze global military strength, compare nations, and simulate strategic alliances using interactive analytics tools.

This project showcases skills in data engineering, analytics, visualization, and software integration suitable for practical data science and analytics applications.